1. Introduction to Natural Language Processing

Natural language processing, usually shortened to NLP, is the field of computer science that teaches machines to read, understand and produce human language. Every day people write emails, send messages, post reviews and search the web, and nearly all of this data is unstructured text. NLP gives us the tools to turn that text into something a program can count, compare and learn from.

The field sits between linguistics, computer science and machine learning. Linguistics explains how words, sentences and meaning are built. Computer science provides the data structures and algorithms to process millions of documents. Machine learning supplies the statistical methods that let a system improve from examples instead of following only hand written rules.

A typical NLP project starts with raw text, cleans it, splits it into tokens, and then represents the tokens as numbers. After that a model can classify the text, translate it, summarise it or answer questions about it. Each of these stages can go wrong in its own way, which is why a careful student of language technology learns to inspect the data at every step.

2. A Short History of NLP

The first serious attempts at machine translation began in the 1950s. In 1954 a public demonstration translated more than sixty Russian sentences into English, and many researchers believed the problem would be solved within a few years. It was not. Language turned out to be far more ambiguous and irregular than early systems assumed.

During the 1960s and 1970s most systems relied on hand written grammar rules. Programs such as ELIZA imitated a conversation by matching keywords, and they showed how easily people read intelligence into simple text patterns. Rule based systems were precise on narrow topics but broke down as soon as the vocabulary grew.

The 1990s brought a statistical revolution. Large collections of text became available, and researchers began to count words and word sequences instead of writing every rule by hand. Hidden Markov models and n-gram language models became standard tools. In the 2010s neural networks took over, first with word embeddings and then with transformer models trained on billions of words.

3. Text Preprocessing

Raw text is messy. It contains capital letters, punctuation, web addresses such as https://www.nltk.org and https://spacy.io, numbers like 2024 or 95%, and stray symbols. Before any model sees the text it is usually cleaned. Common steps are lowercasing, removing noise with regular expressions, splitting the text into sentences and tokens, and removing very common words called stop words.

Tokenization splits a sentence into the units a model will work with. A simple tokenizer splits on spaces, while a better one handles contractions such as don't, abbreviations such as Dr. and hyphenated words. Stemming then cuts words down to a crude root, so that running and runs both become run, while lemmatization uses a dictionary to return the proper base form, so that better becomes good.

Every cleaning step removes some information. Lowercasing makes Apple the company look like apple the fruit. Removing stop words can delete the word not, which turns a complaint into praise. Good preprocessing is therefore a decision made for each task, not a fixed recipe. How much cleaning is right for a search engine? Probably more than for a chatbot.

4. Language Models

A language model assigns a probability to a sequence of words. Given the words the cat sat on the, a good model should think that mat is more likely than refrigerator. This simple idea powers spelling correction, speech recognition, machine translation and predictive keyboards.

The oldest practical language models are n-gram models. A bigram model estimates the probability of a word from the one word before it, and a trigram model from the two words before it. The counts come directly from a training corpus, which makes the model easy to build but also means that any pair of words never seen in training receives a probability of zero. Smoothing methods such as add one smoothing fix this by moving a little probability to unseen events.

Modern language models are neural networks. They read the text as a sequence of tokens, learn a vector for each token and predict the next token using context from the whole sentence. Perplexity is the standard way to compare language models: the lower the perplexity, the less surprised the model is by the test text.

5. Applications of NLP

Search engines use NLP to understand what a query means and which documents answer it. Email services use it to detect spam, and companies use sentiment analysis to learn whether customers are happy with a product by reading thousands of reviews in a few seconds.

Machine translation systems now translate between hundreds of language pairs. Virtual assistants turn speech into text, work out what the user wants, and speak an answer back. In medicine, NLP extracts diagnoses and medicines from doctors' notes so that researchers can study patients at a scale that was impossible by hand.

In education, automatic tools check grammar, grade short answers and suggest reading material. In law and finance, NLP helps professionals search long contracts and news feeds. Almost every one of these applications combines the same basic ingredients: clean text, a good representation and a model trained on enough examples.

6. Challenges and Ethics

Human language is full of ambiguity. The sentence I saw her duck can mean two different things, and the word bank can mean a river bank or a place that keeps money. Sarcasm, slang, spelling mistakes and code switching between languages make the task harder again. Languages with few digital texts, including many South Asian languages, have far fewer tools than English.

Models learn from the text they are given, so they also learn its biases. A system trained on biased data can repeat stereotypes about gender, religion or nationality. Privacy is another concern, because personal messages and medical notes must be handled with great care. Engineers have a responsibility to test their systems on different groups of people before they are used.

Evaluation is also difficult. Accuracy numbers can hide serious mistakes, and a fluent sentence is not always a true one. Researchers therefore combine automatic scores with human judgement, and they publish the limits of their data together with their results.

7. Conclusion

Natural language processing turns the written word into data that computers can use. It begins with careful preprocessing, moves through tokens, counts and probabilities, and ends in applications that millions of people use every day. The techniques in this document, from regular expressions to language models, are the foundations on which larger systems are built.

A student who understands why each step is needed, and what it can damage, is better prepared to build reliable language technology than one who only copies a recipe.